CHIRAG JANBANDHU

Aspiring ML Engineer &Quantitative Finance
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PROJECTS

FraudSentinel: Anomaly Detection Jul 2026– ONGOING

- Analyzed 203K+ Bitcoin transactions and associated transaction network for illicit activity detection.
- Performed comprehensive EDA covering 150+ transaction features, temporal distributions, and severe class imbalance.
- Constructed and validated a directed transaction graph for downstream Graph Neural Network training.
- Discovered 2.4× higher illicit-node concentration among neighboring transactions, providing empirical justification for graph-based learning.
- Built a modular research pipeline for upcoming comparison of XGBoost, GraphSAGE, GCN, and GAT models.

Time Series Forecasting for Portfolio Management Mar 2026 – July 2025

Python · TensorFlow · LSTM · ARIMA · Pandas · NumPy · Scikit-learn · Plotly

- Built an end-to-end quantitative finance pipeline using ARIMA and LSTM models to forecast TSLA stock prices from historical market data.
- Processed and analyzed multi-year daily price data for TSLA, SPY, and BND, performing EDA, volatility analysis, Sharpe Ratio, and Value-at-Risk (VaR) calculations.
- Developed a portfolio optimization engine using Modern Portfolio Theory (Efficient Frontier) to maximize risk-adjusted returns based on forecasted asset performance.
- Implemented Monte Carlo Dropout to generate 95% confidence intervals for future stock price forecasts and evaluated investment strategies through historical portfolio backtesting.

EXPERIENCE

Member — Finance & Economics Club, IIT Kharagpur Aug 2024 – Present

- Researched equity trends and valuation analysis using revenue data and financial indicators for weekly reports.
- Participated in intra-club case competitions focusing on trading psychology, macroeconomics, and options pricing strategies.
- Practically assessed markets in real-time using NSE data and Trading View indicators; applied financial modelling techniques weekly.

Executive Member — Public Policy and Governance Society Aug 2024 – Present

- Assisted in organising policy-focused events, promoting governance-related initiatives, and supporting strategic and operational activities.

SKILLS

Languages & Tools: Python, C, C++, SQL, Git, Jupyter, Excel, Trading View, Anaconda

ML / AI Libraries: PyTorch, PyTorch Geometric, scikit-learn, XGBoost, LightGBM, FAISS, MLflow, VADER, GraphSAGE, CLIP, Captum, FastAPI

Quantitative: Financial Modelling, Portfolio Analysis, Derivatives Pricing, Market Microstructure, Options Pricing Strategies

Coursework: Quantitative Finance (Yale), Financial Derivatives, Corporate Finance, Probability & Statistics, Linear Algebra

EDUCATION

Indian Institute of Technology Kharagpur — B.Sc. Applied Geology Jul 2024 – Ongoing

Cumulative CGPA: 8.39/10.00

Gondia Public School, Gondia — Class XII CBSE May 2021 – Jul 2023

Percentage: 91%

Jawahar Navodaya Vidhyalay, Gondia — Class X CBSE Percentage: 97% Completed 2021

Extracurriculars

- State-level cricket player
- Blood donation camp organizer